Samreen Zafar
23122015

Predicting Rain

Date _____ 20___
M T W T F S S

	Outlook	Tempreture	Hot	Windy	Rain
1	Sunny	Hot	High	weak	NO
2	Sunny	Mild	High	strong	NO
	Cloudy	Hot	High	weak	Yess
		Mild	high	weak	Yess
		Cold	Normal	weak	Yess
	Rainy	Cool	Normal	strong	NO
	cloudy	Mild	Normal	strong	Yess
8	Sunny	Cool	Normal	weak	NO

(Rain) Total = 8 Yess (4) NO (4)

Entropy calculation:

$$= \left[-\frac{4}{8_2} \log(2) \frac{4}{8} \right] - \left[\frac{4}{8} \log(2) \frac{4}{8} \right]$$

$$= -0.5 - 0.5$$

$$HS = -1$$

Calculate Entropy for Out look

	Total	Yess	NO	Entropy
Sunny	3	0	3	0.00
Rainy	3	2	1	0.962
Cloudy	2	2	0	0.00

Calculate entropy for Rainy

$$= \left[\frac{2}{3} \log(2) \frac{2}{3} \right] - \left[\frac{1}{3} \log(2) \frac{1}{3} \right]$$

Date _____ 20___
M T W T F S S

$$= \left[\frac{-0.6 \times \log(0.6)}{\log(2)}\right] - \left[\frac{0.33 \times \log(0.33)}{\log(2)}\right]$$

$$= 0.438 - (-0.524)$$

$$= 0.438 + 0.524$$

$$= -0.962$$

Weighted Entropy $H(S|OT)$

$$H(S|OT) = \frac{3}{8}(0.00) + \frac{3}{8}(0.962) + \frac{2}{8}(0.00)$$

$$= 0.375(0.00) + 0.375(0.962) + 0.25(0.00)$$

$$H(S|OT) = 0.360$$

Gain Info: (IG)

$$IG = HS - H(S|OT)$$

$$= -I - 0.360$$

$$= 0.64$$

Calculate Entropy for Temp:

	Total	Yes	No	Entropy
Hot	2	1	1	0
Mild	3	2	1	0.957
Cool	3	1	2	0.081

Date _____ 20___
M T W T F S S

Calculate Entropy for Hot

$$= -\frac{1}{2}\log(2)\frac{1}{2} - \frac{1}{2}\log(2)\frac{1}{2}$$

$$= -0.5 \times \frac{\log(0.5)}{\log(2)} - 0.5\,\frac{\log(0.5)}{\log(2)}$$

$$= 0.5 + (0.5)$$

$$= 0$$

Calculate Entropy for Mild

$$= -\frac{2}{3}\log(2)\frac{2}{3} - \frac{1}{3}\log(2)\frac{1}{3}$$

$$= -0.6 \times \frac{\log(0.6)}{\log(2)} - 0.3 \times \frac{\log(0.3)}{\log(2)}$$

$$= 0.438 - (-0.519)$$

$$= 0.438 + 0.519$$

$$= 0.957$$

Calculate Entropy for Cool

$$= -\frac{1}{3}\log(2)\frac{1}{3} - \frac{2}{3}\log(2)\frac{2}{3}$$

$$= -(-0.519) - (0.438)$$

$$= 0.519 - 0.438$$

$$= 0.081$$

Date _____ 20___
M T W T F S S

$$= \frac{2}{8}(0) + \frac{3}{8}(0.95) + \frac{3}{8}(0.081)$$

$$= 0 + 0.358 + 0.029$$

$$H(S|T) = 0.387$$

Info gain: $HS - H(S|T)$

$$= 1 - 0.387$$

$$= 0.613$$

Calculate entropy for Hot

	Total	yess	No	Entropy
High	4	2	2	1
Normal	4	2	2	1

High

$$= -\frac{2}{4}\log(2)\frac{2}{4} - \frac{2}{4}\log(2)\frac{2}{4}$$

$$= -0.5 \times \frac{\log(0.5)}{\log(2)} - 0.5 \times \frac{\log(0.5)}{\log(2)}$$

$$= 0.5 + 0.5$$

$$H(S|H) = 1$$

Info gain $= HS - H(S|H)$

$$= 1 - 1$$

$$= 0$$

Date 20
M T W T F S S

Entropy (windy)

	Total	Yess	No	Entropy
weak	5	3	2	0.969
strong	3	1	2	0.916

Entropy for weak

$$= -\frac{3}{5}\log(2)\frac{3}{5} - \frac{2}{5}\log(2)\frac{2}{5}$$

$$= -0.6(-0.736) - 0.4 \times \log\frac{0.4}{\log 2}$$

$$= 0.4416 + 0.528 = 0.96$$

$$= -\frac{1}{3}\log(2)\frac{1}{3} - \frac{2}{3}\log(2)\frac{2}{3}$$

$$= 0.527 + 0.389$$

$$H(S|W) = 0.916$$

Info gain = $HS - H(S/W)$

$= 1 - 0.916$

$= 0.084$

Uni Plus

Date _____ 20____

M T W T F S S

Attribute	WE	Info gain
OL	0.36	1.13
T	0.387	0.613
H	1	0
W	0.916	0.084

